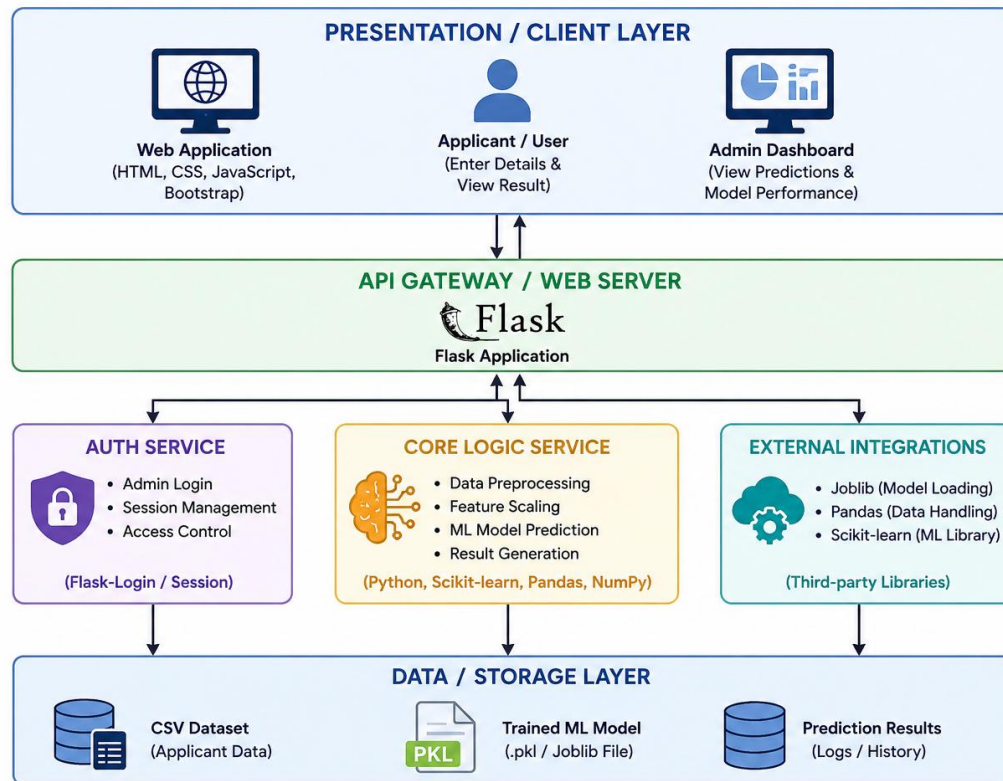


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where applicants or administrators enter applicant details, submit prediction requests, and view prediction results.	HTML5, CSS3, JavaScript, Bootstrap
API Gateway	Receives requests from the frontend, routes them to the backend, validates input, and returns prediction results to the client.	Flask
Core Logic Service <i>(instead of Microservices)</i>	Performs data preprocessing, loads the trained machine learning model, generates approval predictions, and prepares the response displayed on the dashboard.	Python, Scikit-learn, Pandas, NumPy, Joblib
Database / Data Layer	Stores the dataset, trained machine learning model, and prediction-related data required by the application.	CSV Dataset, Joblib (.pkl) Model Files